Department of Computer Engineering
St. Francis Institute of Technology
University of Mumbai
2026-2027

A Mini-Project Report On

“Marathi Sentiment and Emotion Detection System of YouTube Videos”

Subject - Natural Language Processing [CSDC7013]

Under the guidance of

Dr. Nazneen Ansari

By

Swayam Patil (Roll No. 28)
Alston Pereira (Roll No. 30)
Darren Pereira (Roll No. 31)
Shivam Yadav (Roll No. 46)

CERTIFICATE

This is to certify that the Mini-Project entitled “Marathi Sentiment and Emotion Detection System of YouTube Videos” has been carried out by Swayam Patil (Roll No. 28), Alston Pereira (Roll No. 30), Darren Pereira (Roll No. 31), and Shivam Yadav (Roll No. 46) under the guidance of Dr. Nazneen Ansari, in partial fulfilment of the requirement for the subject Natural Language Processing [CSDC7013], during the academic year 2026-2027.

Dr. Nazneen Ansari
Guide

Head of Department
Department of Computer Engineering

Principal
St. Francis Institute of Technology

Table of Contents

Abstract..... i
List of figures..... ii
List of tables.....iii
List of abbreviations..... iv

Chapter 1: Introduction.....1
1.1 Introduction.....1
1.2 Problem definition.....2
1.3 Prerequisites.....3
2. Literature review / related work (20 papers).....5
3. Proposed work / methodology..... 8
3.1 System architecture.....8
3.2 Dataset description, source, exploration, loading and EDA.....9
3.3 Algorithms..... 10
4. Implementation (steps / procedure).....11
5. Experimental setup..... 13
5.1 Parameters.....13
5.2 Tools..... 13
5.3 Libraries.....14
5.4 Framework.....14
5.5 Languages..... 14
6. Evaluation and result / discussion.....15
7. Graphical user interface.....17
8. Code..... 18
Libraries and functions used.....18
9. Conclusion.....20
10. References.....21

Abstract

YouTube contains a large and continuously changing collection of public comments written in Devanagari Marathi, Roman Marathi, Marathi-English code-mixed text, emojis, and short informal expressions. These comments contain useful audience signals, but they are noisy, multilingual, repetitive, and difficult to analyse manually. This project presents an end-to-end Natural Language Processing system that accepts a public YouTube video URL, retrieves accessible comments and replies through the official YouTube Data API, removes duplicate comment IDs, detects Marathi-language signals, and classifies the comments by sentiment and emotion.

The system combines a transparent lexicon and rule-based baseline with lightweight academic model comparison using TF-IDF features and Multinomial Naive Bayes. Sentiment classes are Positive, Negative, and Neutral; the emotion layer produces interpretable labels such as Happy, Sad, Angry, Fear, Surprise, and Neutral. A balanced real sample of 1,500 Marathi examples derived from the L3Cube-MahaSent-MD training data is used for a reproducible holdout benchmark, while the actual YouTube comments remain separate from the benchmark corpus. This distinction is important because a Marathi social-text dataset is not identical to a YouTube-comment dataset.

The project is implemented as the Marathi Lens web application. It displays the YouTube-reported comment total, unique accessible comments fetched, Marathi-signal comments analysed, comments outside the signal, sentiment balance, emotion fingerprint, repeated words, model metrics, a confusion matrix, and downloadable CSV/JSON reports. In a verified scan of video QxeELlJoPvQ, YouTube reported 709 comments and 685 unique accessible comments were returned; 226 comments were identified as Marathi-signal comments. The system demonstrates how classical NLP methods can provide an explainable and practical solution for a subject-level minor project while leaving a clear path for future improvement using transformer models and a manually labelled YouTube test set.

List of Figures

Fig 3.1 System Architecture Diagram..... 8
Fig 3.2 Data Collection and NLP Processing Pipeline.....9
Fig 6.1 NLP Model Comparison and Output Flow.....15
Fig 6.2 Verified Live Scan Comment Count Comparison.....16
Fig 7.1 Marathi Lens Graphical User Interface.....17
Fig 7.2 Annotation and Report Export Panel.....17

List of Tables

Table 2.1 Summary of Literature Reviewed.....5
Table 3.1 YouTube Comment Data Attributes..... 9
Table 3.2 Derived NLP Features.....10
Table 5.1 System Parameters.....13
Table 6.1 Model-wise Performance Comparison.....15

List of Abbreviations

NLP Natural Language Processing

API Application Programming Interface

EDA Exploratory Data Analysis

TF-IDF Term Frequency–Inverse Document Frequency

NB Naive Bayes

ML Machine Learning

POS Part of Speech

WSD Word Sense Disambiguation

CSV Comma Separated Values

JSON JavaScript Object Notation

GUI Graphical User Interface

URL Uniform Resource Locator

F1 F1-Score (harmonic mean of precision and recall)

OCR Optical Character Recognition

LLM Large Language Model

Chapter 1: Introduction

1.1 Introduction

Natural Language Processing is concerned with enabling computers to work with human language. Marathi presents a particularly interesting NLP setting because comments may appear in Devanagari script, Roman transliteration, English, Marathi-English code-mixing, abbreviations, emojis, repeated characters, and informal spelling. YouTube comments add another layer of difficulty: they are short, conversational, asynchronous, and often contain replies, timestamps, links, or non-linguistic noise.

This project builds a complete Marathi YouTube sentiment and emotion detection pipeline. The user provides a public YouTube URL instead of preparing a CSV file. The server extracts the video identifier, requests public metadata and comment threads through the official YouTube Data API, follows pagination, retrieves replies when available, and deduplicates comments by their unique API IDs. The text is normalized and tokenized, Marathi and Roman-Marathi signals are detected, and an explainable classifier assigns sentiment and emotion labels.

The motivation for this project comes from the need to convert a large public conversation into a compact and understandable summary. A student, content creator, researcher, or channel manager can use the system to identify whether the audience response is positive, negative, neutral, happy, angry, surprised, or emotionally mixed. The dashboard also makes the limitations visible by distinguishing the total reported by YouTube from comments actually returned by the API and from the subset classified as Marathi signal.

1.2 Problem definition

Given a public YouTube video URL, the objective of this project is to:

- Retrieve all accessible public top-level comments and replies through the official YouTube Data API using repeated pagination.
- Remove duplicate records created when replies appear both inline in a comment thread and through the reply endpoint.
- Normalize Devanagari, Roman Marathi, Marathi-English code-mixed text, punctuation, emojis, and repeated characters.
- Detect Marathi-language signals using Devanagari and Roman-Marathi markers while reporting comments outside the signal separately.
- Classify comments into Positive, Negative, and Neutral sentiment and assign an interpretable emotion label.
- Aggregate the results into sentiment percentages, emotion frequencies, common words, examples, and downloadable CSV/JSON reports.
- Compare the explainable live baseline with a TF-IDF + Multinomial Naive Bayes benchmark trained on a real Marathi sentiment sample.

The problem is therefore a small but complete NLP system-development problem combining data acquisition, text preprocessing, language identification, sentiment classification, emotion heuristics, model evaluation, web application development, and responsible reporting of API limitations.

1.3 Prerequisites

- Working knowledge of Python or TypeScript and basic data structures.
- Fundamentals of Natural Language Processing, tokenization, stemming, lemmatization, and sentiment analysis.
- Understanding of Marathi script, Roman Marathi, code-mixed language, and common informal YouTube expressions.

• Familiarity with supervised learning, TF-IDF, Naive Bayes, precision, recall, accuracy, macro-F1, and confusion matrices.

• Basic knowledge of JSON APIs, HTTP requests, pagination, and API-key security.

• Basic knowledge of exploratory data analysis and visualisation.

• Familiarity with React, Express, tRPC, and browser-based user interfaces.

• Awareness that benchmark-corpus performance and performance on real YouTube comments are separate experiments.

Chapter 2: Literature Review / Related Work

A wide body of research has explored sentiment analysis, emotion detection, multilingual NLP, code-mixed language, and social-media text classification. The table below summarises twenty relevant papers, datasets, and official technical resources that informed the design choices made in this project—particularly the use of a transparent baseline, TF-IDF with Naive Bayes, Marathi-specific resources, Roman-Marathi handling, and API-aware comment collection.

Sr. No.	Author(s) / Source	Title / Resource	Key Contribution / Relevance
1	Jurafsky & Martin (2009)	Speech and Language Processing	Provides core NLP foundations including language modelling, tagging, parsing, and sentiment-related methods.
2	Manning, Raghavan & Schütze (2008)	Introduction to Information Retrieval	Introduces TF-IDF, vector-space representations, evaluation, and text retrieval concepts.
3	Sebastiani (2002)	Machine Learning in Automated Text Categorization	Establishes classical supervised text classification as a practical approach for document categories.
4	Pang, Lee & Vaithyanathan (2002)	Thumbs up? Sentiment Classification using Machine Learning Techniques	Early comparison of Naive Bayes, maximum entropy, and SVM for sentiment classification.
5	Pang & Lee (2008)	Opinion Mining and Sentiment Analysis	Surveys subjectivity detection, polarity classification, and opinion-mining challenges.
6	Liu (2012)	Sentiment Analysis and Opinion Mining	Provides a broad framework for sentiment lexicons, machine learning, and opinion summarisation.
7	Mohammad (2016)	Sentiment Analysis: Detecting Valence, Emotions, and Other Affectual States	Connects sentiment polarity with emotion and affective language.
8	Devlin et al. (2019)	BERT: Pre-training of Deep Bidirectional Transformers	Demonstrates the value of contextual transformer representations for language understanding.
9	Joulin et al. (2017)	Bag of Tricks for Efficient Text Classification	Shows that simple word representations and linear classifiers can be strong and efficient baselines.
10	Joshi et al. (2021)	L3Cube-MahaSent	Introduces a Marathi social-media sentiment resource with positive, negative, and neutral labels.
11	L3Cube-Pune (2023)	MahaSent-MD: Marathi Multi-Domain Sentiment Dataset	Provides Marathi sentiment examples from multiple domains and supports reproducible benchmarking.
12	L3Cube-Pune (2023)	MeSent: Marathi-English Code-Mixed Sentiment Dataset	Addresses Roman Marathi and code-mixed text, which is common in online comments.
13	L3Cube-Pune (2025)	MahaEmotions	Provides Marathi emotion labels and motivates a separate emotion head rather than treating emotion as identical to sentiment.
14	Kunchukuttan (2018)	Indic NLP Library	Offers Indic-language tokenisation, normalization, transliteration, and script-processing resources.

15	Google Developers (current)	YouTube Data API commentThreads.list	Defines comment-thread pagination, maxResults, nextPageToken, and inline reply behaviour.
16	Google Developers (current)	YouTube Data API comments.list	Defines the endpoint required to retrieve complete reply pages when inline replies are incomplete.
17	Google Developers (current)	YouTube Data API quota documentation	Explains request cost, pagination cost, and the practical limits of large-scale collection.
18	Kowsari et al. (2019)	Text Classification Algorithms: A Survey	Compares classical and deep text classification approaches and their trade-offs.
19	Cambria et al. (2013)	New Avenues in Opinion Mining and Sentiment Analysis	Discusses semantic resources and affective computing for opinion analysis.
20	Bender et al. (2021)	On the Dangers of Stochastic Parrots	Highlights responsible reporting, dataset limitations, and the importance of documenting model scope.

Synthesis of the literature

Three conclusions directly motivate the proposed methodology. First, contextual and transformer models can provide strong language representations, but a small subjectual project benefits from an explainable rule-based baseline and a lightweight TF-IDF + Naive Bayes benchmark that can be demonstrated clearly. Second, Marathi and code-mixed text require language-specific resources; a generic English sentiment model cannot be assumed to work correctly on Devanagari or Roman Marathi. Third, data collection is part of the NLP problem: API pagination, reply retrieval, duplicate removal, deleted comments, and quota costs affect the final dataset. The project therefore reports the YouTube API total, accessible fetched count, Marathi-signal count, and unavailable difference separately.

Chapter 3: Proposed Work / Methodology

3.1 System architecture

The system follows a linear architecture consisting of six stages: URL input, YouTube data acquisition, comment normalization, Marathi signal detection, sentiment/emotion analysis, and result presentation. The same server-side analysis function is used for both a live YouTube URL and the built-in demo mode. The browser never receives the YouTube API key.

- The user enters a public YouTube URL in the Marathi Lens dashboard.
- The server extracts the video ID and requests video metadata and comment threads.
- Pagination continues toward YouTube's reported comment total, subject to a 5,000-comment safety ceiling.
- Replies are obtained from inline thread data and, when necessary, from the comments.list endpoint.
- Comment IDs are stored in a set so duplicate inline and paginated replies are removed.
- The text is cleaned, tokenized, normalised, and passed through Marathi and Roman-Marathi signal heuristics.
- Sentiment, emotion, word counts, evaluation metrics, and downloadable report objects are returned to the dashboard.

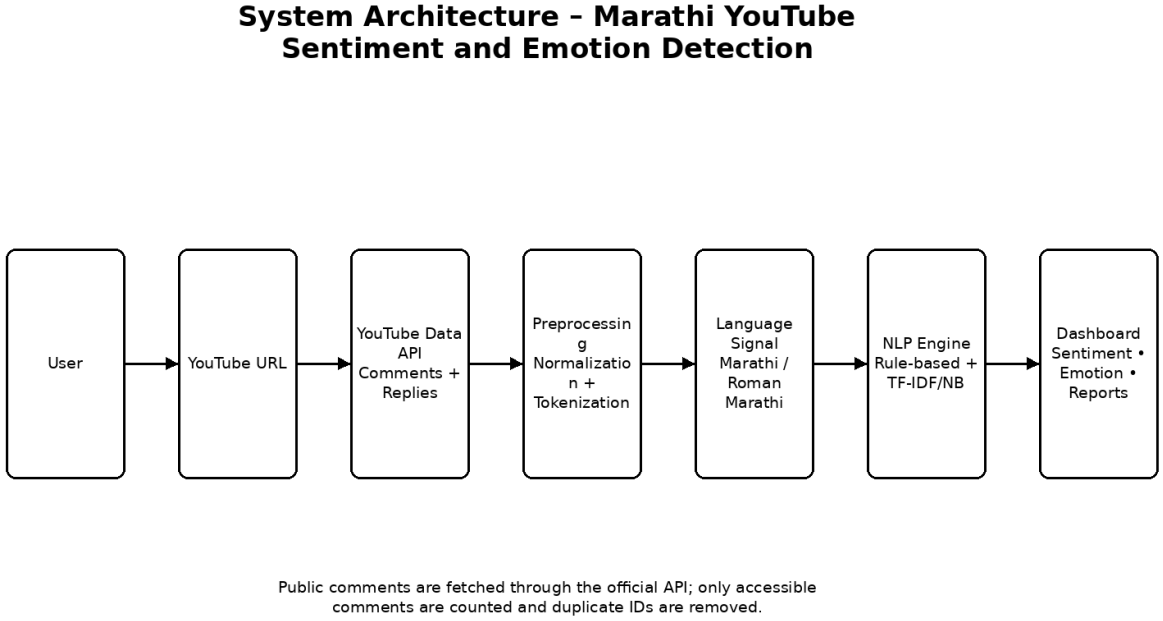


Fig 3.1 System Architecture Diagram

3.2 Dataset description, source, exploration, loading and EDA

The live dataset is not a preloaded CSV. It is generated at run time from public YouTube comments using the official YouTube Data API. This makes the application closer to the requested use case, but it also means that each video has a different number of comments, different language mix, and different amount of noise. For reproducible academic model comparison, the project includes a balanced 1,500-row real sample extracted from the L3Cube-MahaSent-MD training data: 500 positive, 500 negative, and 500 neutral examples. This benchmark sample is not mixed with live YouTube comments and is not used to fabricate comments when a video has fewer comments.

Attribute	Description
video_id	Unique YouTube video identifier extracted from the URL.
video_title / channel_title	Metadata returned by videos.list.
comment_id	Unique identifier used for deduplication.
parent_id	Parent comment identifier for replies.

text	Raw public comment or reply text.
author_display_name	Public display name when returned by the API.
published_at	UTC timestamp returned by YouTube.
like_count	Public like count when available.
total_reply_count	Number of replies reported for a top-level comment.
api_comment_count	YouTube-reported total used as a collection target.

Derived NLP Feature	**Description**
clean_text	Whitespace, punctuation, URL, and repeated-character normalisation.
tokens	Whitespace and Unicode-aware token sequence.
script_signal	Devanagari, Roman Marathi, English, emoji, or mixed indicators.
marathi_signal	Boolean signal used to separate Marathi comments from outside-signal text.
sentiment	Positive, Negative, or Neutral label.
sentiment_score	Signed lexicon score used for an average signal.
emotion	Happy, Sad, Angry, Fear, Surprise, or Neutral.
word_frequency	Repeated non-stopword tokens used in lexical pulse.
source_type	Top-level comment or reply.
is_duplicate	Internal deduplication check based on comment_id.

Exploration and EDA are performed after collection. The dashboard reports the API total, fetched unique comments, Marathi-signal comments, and outside-signal comments. Sentiment proportions are visualised as a donut-style split, while emotion frequencies and repeated tokens provide interpretable qualitative evidence. In the verified QxeELlJoPvQ case study, the API reported 709 comments, 685 unique accessible comments were returned, 226 were identified as Marathi signal, and 459 were outside the signal. The difference between API total and returned count is documented instead of being hidden.

3.3 Algorithms

Rule-based sentiment and emotion baseline

The live analyzer uses Marathi Devanagari markers, Roman-Marathi markers, sentiment lexicons, negation handling, intensity words, and emotion cues. Positive terms such as “छान”, “सुंदर”, “आवडले”, and “खूप” contribute positive evidence; negative terms such as “वाईट”, “कंटाळवाणा”, “नाही”, and “निराश” contribute negative evidence. The classifier is deliberately transparent so that a student can explain why a comment received a label. Emotion is treated as a separate compatibility layer and is not presented as a gold human annotation.

TF-IDF + Multinomial Naive Bayes benchmark

The real Marathi sample is split into 1,200 training rows and 300 holdout rows. Text is represented using TF-IDF word features, and a Multinomial Naive Bayes classifier is evaluated on the holdout set. The dashboard reports accuracy, precision, recall, macro-F1, and a 3 × 3 confusion matrix. The benchmark supports academic comparison but should not be confused with a YouTube-specific gold evaluation set.

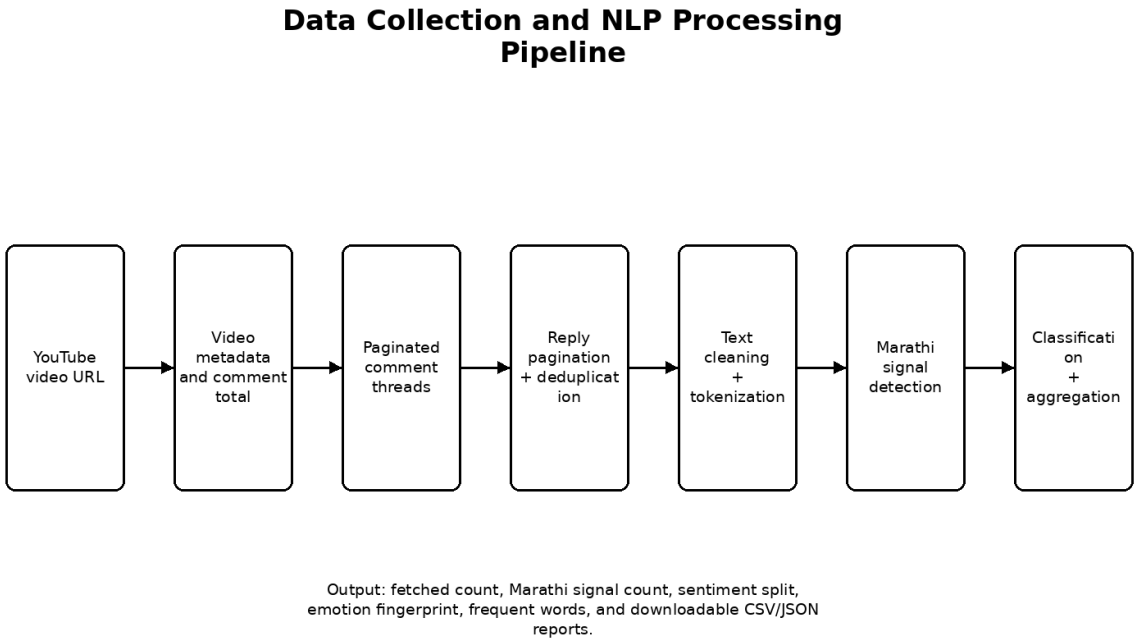


Fig 3.2 Data Collection and NLP Processing Pipeline

Chapter 4: Implementation (Steps / Procedure)

Step 1: Project setup

A full-stack React, Express, tRPC, and TypeScript application named Marathi Lens was created. Server-side environment configuration keeps the YouTube API key away from the browser.

Step 2: URL validation

The client validates that the supplied value contains a supported YouTube URL or video identifier before sending the request. Demo mode does not send a fake live URL to the API.

Step 3: Video metadata acquisition

The server calls videos.list to obtain the title, channel, thumbnail, view count, published date, and YouTube-reported comment total.

Step 4: Comment-thread pagination

commentThreads.list is called with maxResults=100 and nextPageToken. The loop continues toward the reported total or the 5,000 safety limit.

Step 5: Reply retrieval and deduplication

Inline replies are collected. If totalReplyCount indicates more replies, comments.list is called with the parent ID and paginated. All comment IDs are deduplicated.

Step 6: Text preprocessing

The analyzer removes unnecessary whitespace, normalises punctuation and repeated characters, preserves Devanagari, detects Roman-Marathi markers, and tokenises the text.

Step 7: Language signal detection

Devanagari characters, Roman-Marathi vocabulary, Marathi suffix patterns, and code-mixed markers are used to identify the Marathi signal. The remaining comments remain visible as an outside-signal count.

Step 8: Sentiment and emotion classification

The transparent baseline assigns sentiment and emotion labels, calculates an average signed signal, and builds word-frequency and emotion-frequency summaries.

Step 9: Academic benchmark

The 1,500-row real Marathi sample is split into train and holdout partitions for TF-IDF + Naive Bayes comparison metrics.

Step 10: Report generation

The browser creates CSV and JSON downloads containing the video metadata, counts, labels, scores, and comment-level examples. An annotation template supports manual gold labelling of 100–150 YouTube comments.

Step 11: Verification

The project was tested with unit tests, TypeScript compilation, production build, Demo mode, and live videos. The QxeELlJoPvQ case study was used to verify full pagination and duplicate removal.

Chapter 5: Experimental Setup

5.1 Parameters

Component	Key Parameters / Configuration
YouTube collection	maxResults=100 per page; target uses API commentCount; safety maximum=5,000.
Reply collection	Inline replies plus comments.list pagination by parentId.
Deduplication	Set of unique YouTube comment IDs.
Rule baseline	Marathi lexicon, Roman-Marathi markers, negation and intensity cues.
Real benchmark	1,500 rows; 1,200 train and 300 holdout; 500 per sentiment class.
TF-IDF + NB	Word-level TF-IDF with Multinomial Naive Bayes.
Metrics	Accuracy, precision, recall, macro-F1, and confusion matrix.
Live case study	QxeELlJoPvQ: API total 709; unique accessible fetched 685.

5.2 Tools

- Visual Studio Code or browser-based WebDev environment for implementation and debugging.
- Git and project checkpoints for version control and rollback.
- Official YouTube Data API v3 for video and comment retrieval.
- Browser preview for live end-to-end verification.
- Python utilities for dataset inspection and report generation.

5.3 Libraries

- React, React DOM, Wouter, and Lucide React for the client interface.
- Express and tRPC for typed server procedures and API communication.
- Zod for input validation.
- TypeScript for implementation and type safety.
- Vitest for unit tests.
- TF-IDF and Multinomial Naive Bayes utilities for the benchmark.
- Python-docx and LibreOffice for report generation and PDF export.

5.4 Framework

The project uses a React + Express + tRPC full-stack framework. The framework was selected because it provides a single typed contract between the browser and the server, while allowing the server-side YouTube key and collection logic to remain private. The dashboard is designed for a student project rather than a large-scale production analytics platform.

5.5 Languages

TypeScript is used for the frontend and server-side analysis engine. Python is used for dataset inspection and report automation. Marathi Devanagari and Roman Marathi are the primary language forms analysed, while English words and emojis are retained as part of the code-mixed comment environment.

Chapter 6: Evaluation and Result / Discussion

6.1 Benchmark model results

The TF-IDF + Multinomial Naive Bayes comparison was evaluated on a 300-row holdout split from the balanced real Marathi sample. The current dashboard reports 60% accuracy, 60% precision, 60% recall, and 60% macro-F1. These values are benchmark values for the selected sample and should not be reported as the accuracy of the live YouTube comments, because the YouTube comments have not yet received manual gold labels.

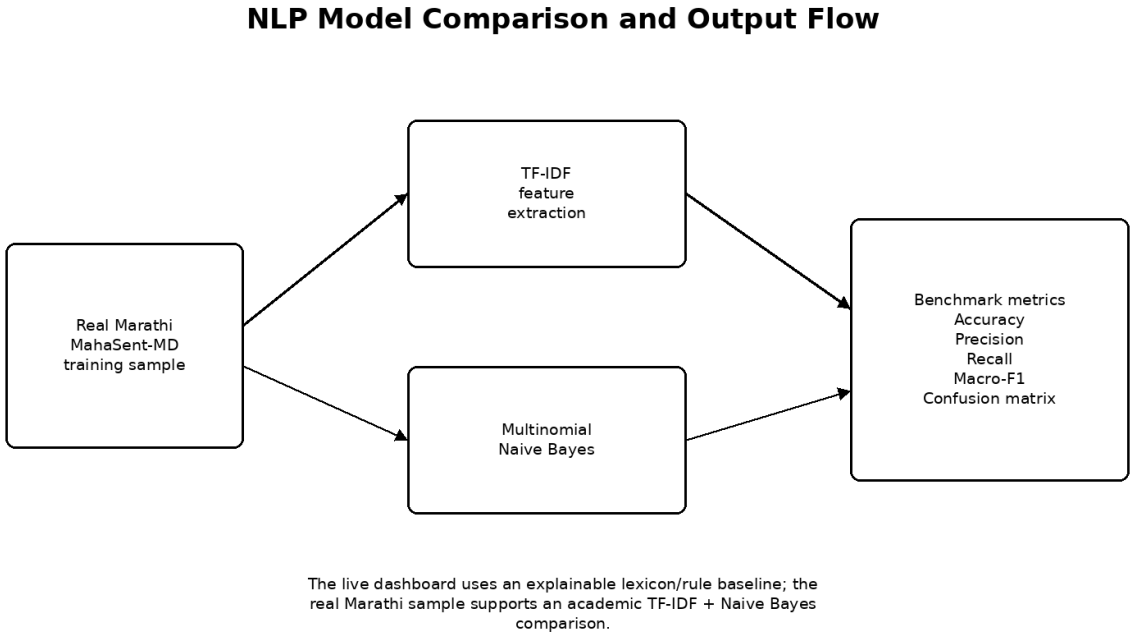


Fig 6.1 NLP Model Comparison and Output Flow

6.2 Live retrieval results

The full-pagination collector was verified with the video QxeELlJoPvQ. YouTube reported 709 comments. The system returned 685 unique accessible comments after combining top-level comments and replies and removing duplicate IDs. The result contained 226 Marathi-signal comments and 459 comments outside the signal. The difference of approximately 24 comments is reported as unavailable or not returned by the public API; the system does not fabricate replacements.

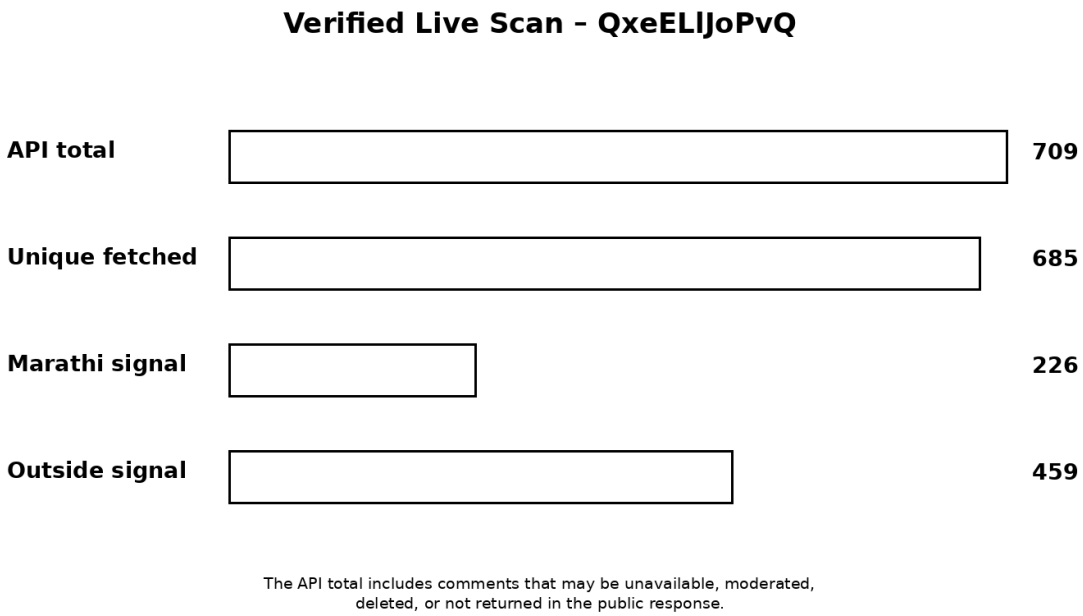


Fig 6.2 Verified Live Scan Comment Count Comparison

6.3 Sentiment and emotion results

For the verified live scan, the Marathi-signal subset was classified as 19% positive, 3% negative, and 78% neutral. The average signed signal was approximately +0.46, indicating a mildly positive direction despite a neutral majority. Neutral was the dominant emotion label in the current rule-based output. This illustrates why a system should show both distributions and example comments rather than relying on only one headline label.

6.4 Summary table

Task / Component	Result
Collection	685 unique accessible comments returned from a YouTube-reported total of 709 in the verified case study.
Language signal	226 Marathi-signal comments and 459 outside-signal comments in the verified case study.
Sentiment baseline	19% positive, 3% negative, and 78% neutral for the Marathi-signal subset in the verified case study.
Benchmark classifier	TF-IDF + Multinomial Naive Bayes: 60% accuracy, precision, recall, and macro-F1 on the selected holdout sample.
Reporting	CSV, JSON, annotation template, word frequency, emotion fingerprint, and confusion matrix outputs.

6.5 Visual insights

The dashboard visualises sentiment balance, emotion frequencies, lexical pulse, fetched-versus-API totals, Marathi-signal versus outside-signal counts, and representative comments. These views consistently show that YouTube comments are noisier than curated benchmark examples. Long emoji strings, repeated characters, timestamps, short replies, and Roman-English fragments can be returned by the API but may not contain reliable Marathi sentiment evidence. The separate counts make this behaviour visible and support a more honest academic discussion.

Chapter 7: Graphical User Interface

The finished system is exposed through the Marathi Lens web application with a wide-screen dashboard and three primary navigation areas: Overview, Comments, and Language signals.

7.1 Analysis workspace

The Overview screen provides a URL field, Analyze comments button, Demo button, system status, and explanatory helper text. The user does not upload a CSV. The server key remains private, and the collection note states that the system fetches available comments up to the safety ceiling.

7.2 Result and language-signal panel

After analysis, the application shows the video title, thumbnail, channel, view count, comments fetched, Marathi-signal count, unavailable difference, sentiment balance, average signal, emotion fingerprint, and repeated words. A dedicated language-signal card separates Devanagari Marathi, Roman Marathi / mixed text, and other/noisy comments. A preprocessing trace shows original text, cleaned text, and tokens for representative examples. Comments outside the Marathi signal are reported but do not silently inflate the Marathi sentiment chart.

7.3 All-comment browser and report export

The Comments workspace preserves every unique accessible comment returned by the API, not only the Marathi subset used for sentiment aggregation. Users can search comment text or author name, filter by language, sentiment, and emotion, sort by recency, likes, or sentiment score, and browse ten comments per page. The results view also includes the real Marathi benchmark metrics, a confusion matrix, and a short explanation of the benchmark-versus-YouTube distinction. The user can download a CSV comments file, a JSON report, and an annotation template. The template allows 100–150 real comments to be manually labelled with gold sentiment, gold emotion, and notes for a final YouTube-specific evaluation.

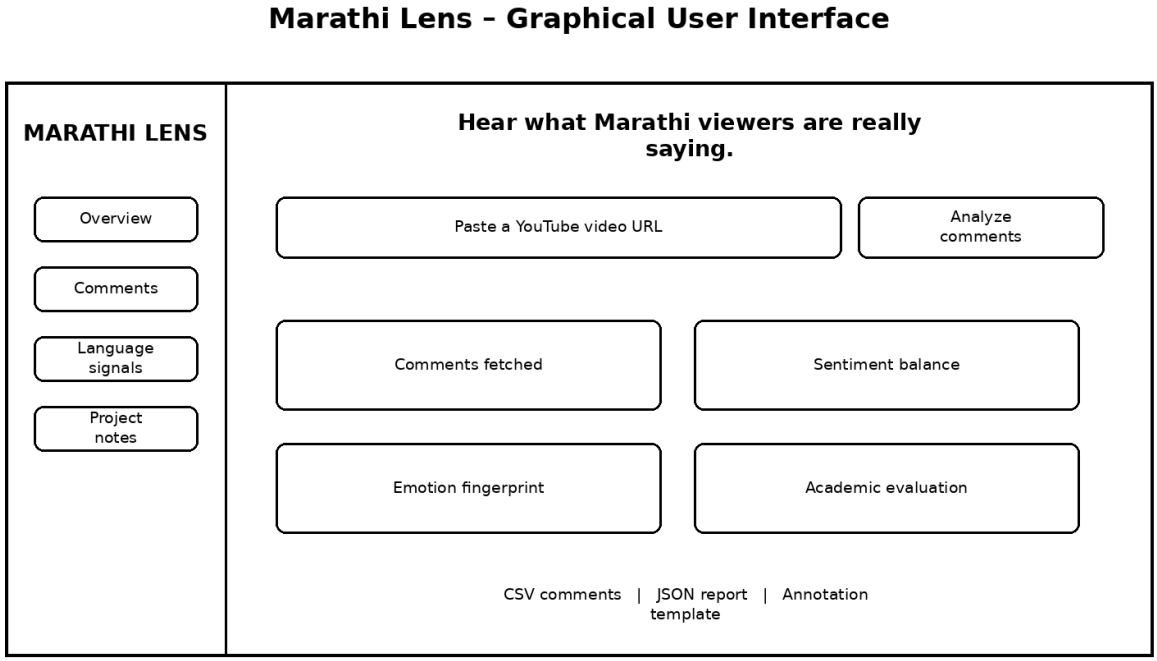


Fig 7.1 Marathi Lens Graphical User Interface

Fig 7.2 Annotation and Report Export Panel

Annotation template | CSV comments | JSON report

The panel is intentionally simple. It allows the student to save a scan for the project report, export comment-level outputs, and create a manually labelled evaluation set. The interface does not claim that a benchmark corpus is a substitute for gold labels from the target YouTube domain.

Chapter 8: Code

The core implementation is organised into a typed server analysis module, a tRPC router, a React page, a real Marathi benchmark dataset module, and automated tests. The following simplified excerpt reproduces the main collection and deduplication logic; UI and API error-handling code is summarised for brevity.

```
const reportedCommentCount = Number(video.statistics?.commentCount ?? 0);
const collectionTarget = Math.min(maxComments, Math.max(1, reportedCommentCount || maxComments));
const comments: RawComment[] = [];
const seenCommentIds = new Set<string>();

const addComment = (comment: RawComment) => {
  if (seenCommentIds.has(comment.id) || comments.length >= collectionTarget) return;
  seenCommentIds.add(comment.id);
  comments.push(comment);
};

while (comments.length < collectionTarget) {
  const page = await fetchCommentThreads(videoId, pageToken);
  for (const thread of page.items) {
    addComment(toRawComment(thread.snippet.topLevelComment));
    await fetchAllReplies(thread, addComment);
  }
  if (!page.nextPageToken) break;
  pageToken = page.nextPageToken;
}
```

Libraries and functions used

Library / Function	Purpose in this project
extractYouTubeVideoId	Accepts watch, short, embed, and youtu.be URLs and returns a video ID.
videos.list	Loads title, channel, thumbnail, views, date, and API comment total.
commentThreads.list	Loads paginated top-level comment threads with inline replies.
comments.list	Loads additional reply pages by parent comment ID.
Set<string>	Deduplicates top-level comments and replies by comment ID.
cleanText / tokenize	Normalises text and returns Unicode-aware tokens.
Marathi signal heuristics	Detects Devanagari, Roman-Marathi, and code-mixed indicators.
analyzeComment	Assigns sentiment, score, and emotion using transparent cues.
buildModelComparison	Trains/evaluates TF-IDF + Naive Bayes on the real sample.
CSV / JSON download helpers	Exports comment-level and summary report data.
Vitest	Tests URL extraction, sentiment, demo results, dataset size, metrics, language categories, and all-comment preservation.

Chapter 9: Conclusion

This project demonstrates a complete, end-to-end Natural Language Processing workflow for turning a public YouTube video into an interpretable Marathi audience signal. The system retrieves comments and replies directly from YouTube, follows pagination, removes duplicate IDs, detects Marathi and Roman-Marathi signals, classifies sentiment and emotion, and presents the result through a browser dashboard. It avoids the unrealistic assumption that every video must be represented by a fixed CSV file or exactly 100–150 comments.

The results confirm several practical observations. First, the number shown in a sentiment chart can be smaller than the number of comments on YouTube because the chart is based on the Marathi-signal subset. Second, the YouTube API total is not always identical to the number of accessible comments returned; deleted, moderated, hidden, or otherwise unavailable items must be reported rather than invented. Third, a transparent rule-based baseline is suitable for a minor project because the student can explain the cues, while TF-IDF + Naive Bayes provides a conventional academic benchmark using a real Marathi corpus.

The project also highlights important limitations. The live rule-based sentiment and emotion labels have not been validated against a manually annotated YouTube gold set. Marathi benchmark corpora are not identical to YouTube comments, and the distribution of emojis, spam, code-mixing, sarcasm, and very short replies can change the results. YouTube’s reported total may be larger than the accessible response because comments may be deleted, moderated, hidden, or unavailable to the API. A test-case checklist covers valid URLs, invalid URLs, disabled comments, replies, Roman Marathi, English text, and empty input. Future improvements include collecting and manually labelling 100–150 comments from multiple videos, adding MeSent for stronger code-mixed robustness, using the human-labelled splits of MahaEmotions for emotion evaluation, comparing a Marathi transformer model, and supporting cached multi-video trend analysis.

Chapter 10: References

1. Jurafsky, D., & Martin, J. H. Speech and Language Processing, 2nd edition, Prentice Hall, 2009.

2. Manning, C. D., Raghavan, P., & Schütze, H. Introduction to Information Retrieval. Cambridge University Press, 2008.

3. Sebastiani, F. Machine Learning in Automated Text Categorization. ACM Computing Surveys, 34(1), 2002.

4. Pang, B., Lee, L., & Vaithyanathan, S. Thumbs up? Sentiment Classification using Machine Learning Techniques. EMNLP, 2002.

5. Pang, B., & Lee, L. Opinion Mining and Sentiment Analysis. Foundations and Trends in Information Retrieval, 2(1–2), 2008.

6. Liu, B. Sentiment Analysis and Opinion Mining. Morgan & Claypool, 2012.

7. Mohammad, S. M. Sentiment Analysis: Detecting Valence, Emotions, and Other Affectual States. In Emotion Measurement, 2016.

8. Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. NAACL, 2019.

9. Joulin, A., Grave, E., Bojanowski, P., & Mikolov, T. Bag of Tricks for Efficient Text Classification. EACL, 2017.

10. Joshi, A., et al. L3Cube-MahaSent: A Marathi Social Media Sentiment Dataset. WASSA / ACL Workshop, 2021.

11. L3Cube-Pune. MahaSent-MD: Marathi Multi-Domain Sentiment Dataset. PACLIC, 2023. https://aclanthology.org/2023.paclic-1.28/

12. L3Cube-Pune. MeSent: Marathi-English Code-Mixed Sentiment Dataset. 2023. https://arxiv.org/abs/2306.14030

13. L3Cube-Pune. MahaEmotions: Marathi Emotion Dataset. 2025. https://arxiv.org/html/2506.00863v2

14. Kunchukuttan, A. Indic NLP Library and Indic NLP Resources. GitHub repositories, accessed 2026.

15. Google Developers. YouTube Data API: commentThreads.list. https://developers.google.com/youtube/v3/docs/commentThreads/list

16. Google Developers. YouTube Data API: comments.list. https://developers.google.com/youtube/v3/docs/comments/list

17. Google Developers. YouTube Data API Quota Calculator and Quota Costs. https://developers.google.com/youtube/v3/determine_quota_cost

18. Kowsari, K., Jafari Meimandi, K., Heidarysafa, M., Mendu, S., Barnes, L., & Brown, D. Text Classification Algorithms: A Survey. Information, 10(4), 2019.

19. Cambria, E., Schuller, B., Xia, Y., & Havasi, C. New Avenues in Opinion Mining and Sentiment Analysis. IEEE Intelligent Systems, 28(2), 2013.

20. Bender, E. M., Gebru, T., McMillan-Major, A., & Shmitchell, S. On the Dangers of Stochastic Parrots. FAccT, 2021.

21. L3Cube-Pune. MarathiNLP repository and dataset documentation. https://github.com/l3cube-pune/MarathiNLP

22. Google. YouTube Data API v3 documentation and revision history. Accessed 2026.

23. scikit-learn developers. Feature extraction and Naive Bayes documentation. https://scikit-learn.org/

24. Manning, C. D., et al. Stanford Natural Language Processing Group resources for text classification and information retrieval.